Kahawa Roots

Traceability Platform

Track coffee from a Kenyan farm to a UK cup allowing anyone to scan a QR code and view its full journey.

Transparent traceability builds buyer trust and enables premium pricing.

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Pre-Interview Task — Developer Intern — April 2026

Proof of Concept

I've already built a live system on the **same stack** I'm proposing:

DeltaEd — Live on GCP Cloud Run

GCP • React • Python • BigQuery

(Built on a similar stack)

How Kahawa Roots Tracks Coffee, End to End

The Journey in 4 Steps

1. **Data Ingestion** — Farmers or admins submit data via a web form or CSV upload.
2. **Validation & Cleaning** — Every field is automatically validated. Invalid rows are rejected and logged for review.
3. **Secure storage** — Raw data is preserved as an immutable backup. Cleaned data is stored in the Database with timestamps & GPS Coordinates at each stage.
4. **Customer Visibility** — Each shipment is assigned a QR code linking to a public page displaying its journey- map, timeline & farmer details.

FLOW:

Data flows in → Validated → stored → visible via QR.

Non-Technical Data Entry

Designed for non-technical users:

Web Form

- Input one shipment at a time via the admin dashboard.
- No technical knowledge needed.

CSV/Excel Upload

- Drag-and-drop bulk upload.
- Automatic row-level validation.

Both methods feed the same pipeline, ensuring constant validation and storage.

Access: Admins manage data, public users access information via QR.

Technology Choices + Bonus

Tech Stack & Why

Component	Choice & Why
Cloud	**GCP**(London europe-west2), GDPR-compliant, eligible for **$200K startup credits**, scales to zero.
Backend	**Python + REST API** on Cloud Run - reliable.
Frontend	**Next.js is** widely adopted, easy to hire server-side rendering ensures fast QR pages load.
Database	**Cloud SQL (Postgres)** — reliable, free-tier, low latency.
Storage	**Cloud Storage** — cost-effective, immutable raw data backup.
Processing	**Cloud Functions** — event-driven, pay-per-use.
IaC / CI	**Terraform + GitHub Actions** — infra as code with automated deployment.
Maps	**Google Maps API** — native integration with GCP for visualising.

Medallion Architecture

Bronze
Raw CSV/JSON → Cloud Storage

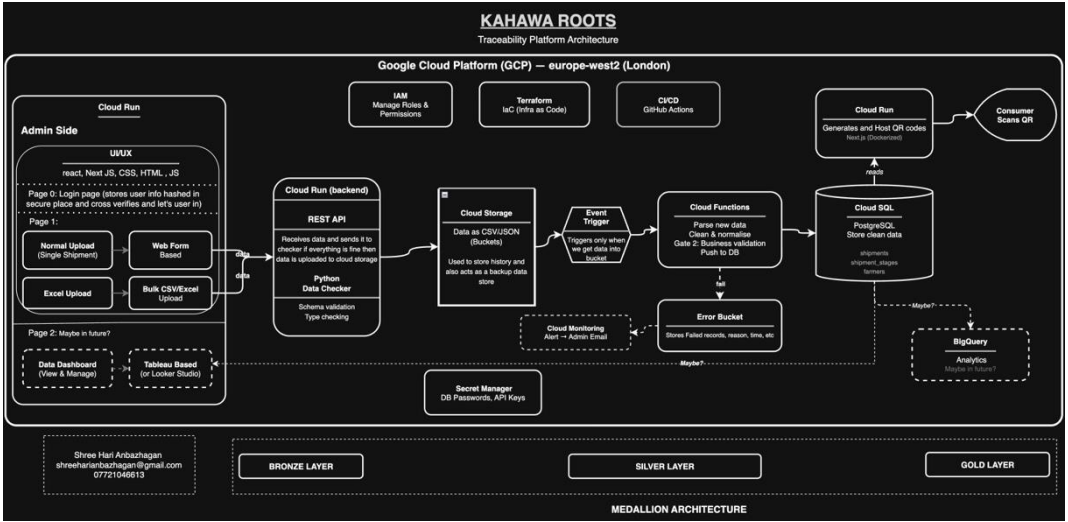
Silver
Cleaned data → PostgreSQL

Gold (future)
QR Pages + BigQuery Analytics

Bonus: Database Schema

```
CREATE TABLE farmers (
  id UUID PK, name VARCHAR, region VARCHAR, farm
  size DECIMAL, lat/lon DECIMAL(9,6), cooperative
  VARCHAR);
CREATE TABLE shipments (
  id UUID PK, batch id VARCHAR UNIQUE, farmer id
  UUID FK→farmers, grade VARCHAR, weight kg
  DECIMAL, status VARCHAR, created at TIMESTAMP
  DEFAULT NOW());
CREATE TABLE shipment stages (
  id UUID PK, shipment id UUID FK→shipments, stage
  VARCHAR, location VARCHAR,
  timestamp TIMESTAMP, notes TEXT);
```

System Architecture



🔗 Click here for full-resolution architecture

Why These Choices Work for Kahawa Roots

£ Cost effective at Small Scale

- **Cloud Run** scales to zero — Pay only for run, no cost when idle. No idle infrastructure costs.
- **Cloud Functions** run only on demand — a few pence per execution, not a monthly server bill.
- **GCP** — eligible for startup credits (up to $200K). The platform could run for months before costing anything.
- **PostgreSQL** — offers a robust, free-tier-friendly solution, no licensing fees.

Easy to Maintain

- **Modular design:** 3 small apps instead of one monolith.
- **Infrastructure as code** — Full system reproducible via configuration.
- **Auto-deploy:** push code tested and deployed instantly.

Safe & Reliable

- **Raw data is never deleted** — every CSV and form submission is backed up untouched. If anything goes wrong, the original data is present as backup.
- **Two-step validation** — data is checked at entry *and* again before it is stored in the database. Bad rows are rejected with clear error messages.
- **Passwords stored securely** — database credentials and API keys stored securely in the vault, and not in code.
- **UK data centre** — runs in London region (europe-west2). Data stays in the UK, GDPR-compliant data handling.

Scales When You're Ready

- **BigQuery** ready — when there's enough data, plug in analytics without rebuilding anything.
- **Dashboard** — Looker Studio for business insights